LESSON 6.10

Divide Mixed Numbers

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can explain what dividing with mixed numbers means, rewrite them as improper fractions, find the quotient with common denominators or by multiplying by the reciprocal, and check it by multiplying.

WHY THIS MATTERS Agent Chen is mapping a suspect's route. The total distance is 3 1/2 miles, and each segment between landmarks is 1/4 mile. She needs to know how many segments are on the route to place surveillance cameras at every landmark.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Mixed Number *Spanish: Número mixto*	A whole number plus a fraction, like 2 1/3.	2 1/3 means 2 wholes and 1/3 more — picture 2 full circles and 1/3 of another
Improper Fraction *Spanish: Fracción impropia*	A fraction where the top is bigger than or equal to the bottom, like 7/4.	7/3 is improper because 7 > 3. It equals 2 1/3 as a mixed number.
Convert *Spanish: Convertir*	To change a number to a new form but keep the same value.	2 1/3 → multiply 2 × 3 = 6, add 1 → 7/3. Same value, different form.
Simplify *Spanish: Simplificar*	To write a fraction with smaller numbers that names the same amount, like 2/4 = 1/2.	8/6: GCF is 2 → 8÷2 / 6÷2 = 4/3 = 1 1/3
Reciprocal *Spanish: Recíproco*	A fraction turned upside down. Dividing by a fraction gives the same answer as multiplying by its reciprocal.	The reciprocal of 3/4 is 4/3. To divide by 3/4, multiply by 4/3.

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Agent Chen has a 3 1/2-mile route cut into 1/4-mile segments. I want 3 1/2 ÷ 1/4.

1. First I convert 3 1/2. Multiply the whole by the bottom: 3 × 2 = 6. Add the top: 6 + 1 = 7. So 3 1/2 = 7/2.
2. Now I ask how many 1/4s fit in 7/2. In fourths, 7/2 = 14/4, and 14 fourths ÷ 1 fourth = 14 — the common-denominator method.
3. The reciprocal shortcut gives the same answer: 7/2 × 4/1 = 28/2 = 14.
4. Check by multiplying: 14 × 1/4 = 14/4 = 3 1/2. So 3 1/2 ÷ 1/4 = 14 segments.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Let's try 2 1/2 ÷ 1/2. First convert 2 1/2: 2 × 2 = 4, plus 1 = 5, so 2 1/2 = 5/2.

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- Before dividing, I convert 2 1/2 to _____ because it is easier to _____ with improper fractions.

Word bank Mixed Number | Improper Fraction | Convert | Simplify | Reciprocal | improper | 5/2 | convert | multiply | reciprocal | The reciprocal shortcut

WATCH OUT!

A common mistake

A common mistake in Divide Mixed Numbers is dividing the whole-number parts and fraction parts separately instead of converting first

LESSON 6.10

Divide Mixed Numbers

Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Convert each mixed number to an improper fraction.

Mixed Number	Improper Fraction
1 3/4	
2 2/3	
4 1/2	
3 1/5	

STEPS TO TRY

1. Every conversion keeps its denominator — 1 3/4 stays in fourths.
2. Use whole × denominator + numerator for each new top number.
3. Row one: 1 × 4 = 4, then add the 3. Put that sum over 4 and move to the next row.

SHOW YOUR WORK

PART 2 Practice the skill

show every step

2 **Circle the letter of the best answer. Show how you know.**

What is 1 1/2 ÷ 3/4?

(A) 1 1/8

(B) 2

(C) 3/4

(D) 9/8

STEPS TO TRY

1. 1 1/2 is a mixed number — it has to become one improper fraction before you divide.
2. Convert: whole × denominator + numerator (1 × 2 + 1), then multiply by the reciprocal with 3/4.
3. 1 1/2 = 3/2. Flip 3/4 to 4/3 and set up 3/2 × 4/3 — multiply across, then simplify.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

What is 2 1/4 ÷ 3/4?

(A) 1 1/2
(B) 2 1/3
(C) 3
(D) 9/16

STEPS TO TRY

1. Convert 2 1/4 to an improper fraction before anything else.
2. 2 × 4 + 1 gives the new numerator over 4. Then flip the divisor 3/4 to 4/3.
3. Set up your improper fraction × 4/3 — the fours cancel, leaving an easy division.

SHOW YOUR WORK

PART 3 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

Convert 3 2/5 to an improper fraction.

(A) 6/5
(B) 15/5
(C) 17/5
(D) 32/5

STEPS TO TRY

1. Converting keeps the denominator 5 the same — only the top number changes.
2. New numerator = whole × denominator + old numerator.
3. Compute 3 × 5 first, then add the 2. Put that sum over 5 and find the match.

Rewrite each fraction _____

Work it out _____

Simplify _____

1. **Same size pieces** *Common denominator first.*
2. **Operate** *Only then add or subtract.*
3. **Simplify** *Divide out common factors.*

EXPLAIN YOUR THINKING

Why is converting to an improper fraction the first step when dividing mixed numbers?

SENTENCE STARTER Before dividing, I convert 2 1/2 to ___ because it is easier to ___ with improper fractions.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help